

UNIVERSITY PARIS 1 PANTHÉON-SORBONNE

MASTER'S THESIS — M1 FINANCE
FMGR Track

**Assessment of the Uncertainty Dynamics Affecting the
Convenience Yield of Oil.**

Gianni Mazaud

**Commodity Markets:
Geopolitics and Risk Premia**

Table of Contents

<i>Introduction</i>	3
<i>Literature Review</i>	5
Options, Futures, and Other Derivatives, J. Hull (2022)	5
Measuring Geopolitical Risk, D. Caldara & M. Iacoviello.	6
The Role of Storage in Commodity Markets: Indirect Inference Based on Grains Data, C. Gouel & N. Legrand.	7
<i>Empirical Study</i>	8
Model Definition.	8
Data Collection.	9
Stationarity of the Data.	10
Final Results.....	11
Analysis of the Standard Error of Residuals	12
ANOVA Analysis	13
<i>Discussion of Results</i>	14
Interpretation of the Final Model.....	14
Decomposition of Explained Variability.....	16
Interpretation of the Dynamics.....	17
Limitations of the Model.	18
Some Avenues for Reflection.....	20
Conclusion.	20
<i>References / Bibliography</i>	21

Introduction

Whatever the scale, from ordinary households to nation-states, oil sits at the center of concerns. As a strategic resource, it drives fuel prices, transportation costs, competitiveness, and so on. At a time when conflicts in the Middle East are fueling a sharp rise in oil and fuel prices, many actors struggle to plan ahead in the face of this geopolitical situation. Commodity markets appear as one of the main transmission channels for these events, reflecting in real time the fears of market participants.

Among commodity markets, the crude oil market occupies a singular position. Given its weight in the global economy, it concentrates the attention of market participants. This period has also seen two structural shifts assert themselves. On the one hand, the rise of the United States as the world's leading producer of crude oil. On the other hand, a growing financialization of oil markets has taken place, with the rise of commodity ETFs, the integration of derivatives markets, and the increasing participation of institutional investors.

Country	Million barrels per day	Share of world total
United States	21.91	22%
Saudi Arabia	11.13	11%
Russia	10.75	11%
Canada	5.76	6%
China	5.26	5%
Iraq	4.42	4%
Brazil	4.28	4%
United Arab Emirates	4.16	4%
Iran	3.99	4%
Kuwait	2.91	3%
Total top 10	74.59	73%
World total	101.81	

Source: "What countries are the top producers and consumers of oil?", 2023 (eia.gov)

The valuation of commodity futures contracts constitutes an important issue in market finance. Unlike pure financial assets, physical commodities, which can be regarded as "consumption" assets, involve storage costs, logistical constraints, and a benefit associated with the physical holding of the resource. This benefit, referred to as the "convenience yield," occupies a fundamental place in the "cost of carry" theory introduced by J. Hull (2022) in *Options, Futures, and Other Derivatives; Risk Management and Financial Institutions*.

In light of these new trends shaping the market and the geopolitical situation, it is worthwhile to identify the role of uncertainty in the convenience yield. Whether financial or geopolitical, uncertainty appears to be a first-order variable in the evolution of oil, but what is truly its effect on the convenience yield?

Can the holding of this consumption asset be valued by overall uncertainty? Or, on the contrary, is it penalized by it?

We will attempt to assess which uncertainty dynamics affect the convenience yield of oil under two forms. On one side, the impact of geopolitical uncertainty; on the other, the impact of economic and financial uncertainty.

To this end, we seek to carry out an econometric modeling of this yield for one-month WTI (West Texas Intermediate) crude oil futures contracts. The objective is to identify the importance of two specific determinants in the evolution of this holding benefit: uncertainty related to geopolitical risk and uncertainty related to economic risk. For this purpose, we will draw on two indices: the Geopolitical Risk (GPR) Index and the Volatility Index (VIX), which measures the implied volatility of financial markets, calculated from S&P 500 index options.

The choice of these variables aims to encompass the various shocks, both geopolitical and economic. The GPR index obviously captures geopolitical shocks, but, as emphasized by D. Caldara and M. Iacoviello, the authors of the study behind this index, it does not react to periods of financial and economic distress or to presidential elections, for example. Adding a volatility index fills precisely this gap. This variable, unlike the GPR index, does not capture the same type of events, such as Russia's annexation of Crimea or the September 11, 2001 attacks in New York.

We will use a data sample covering a 10-year period between 2014 and 2024, a period marked by a succession of shocks of unprecedented scale: the collapse of oil prices between 2014 and 2016 following the shale oil revolution, the Covid-19 pandemic and the episode of negative WTI prices in April 2020, the post-pandemic recovery of 2021, and finally Russia's invasion of Ukraine in February 2022, along with the sanctions and reorganizations of global oil flows that followed.

Regarding the research works that will form the basis of our study, we will rely on the work of J. Hull for the valuation of futures contracts, and more specifically those whose underlying is a consumption asset, i.e., an asset held primarily for the purpose of being consumed. Next, we will study the construction of the GPR index by M. Iacoviello and D. Caldara in order to correctly integrate it into our model while identifying its possible shortcomings. Finally, we will rely on the work of C. Gouvello and N. Legrand to establish the value of storage costs in futures contracts on

consumption assets, with the aim of calculating the value of the convenience yield on the underlying studied over time.

Literature Review

Options, Futures, and Other Derivatives, J. Hull (2022)

A common model for valuing futures contracts is the one proposed by J. Hull. The model's framework rests on the following assumptions:

- Absence of transaction costs.
- Participants are subject to the same tax rate regardless of their profits.
- Participants can borrow and lend money at the same risk-free rate.
- Arbitrage opportunities are seized as soon as they appear.

Since our study focuses on oil markets, we are interested in futures contracts on a consumption asset, defined as "an asset held primarily for consumption purposes." As a result, there is a storage cost, considered a negative income, that directly affects the futures price. There are two ways to incorporate it into the calculation: as proportional to the price (in which case it is a rate) or as a fixed cost set at initiation. In the case of a cost proportional to the price known at contract initiation, it is modeled by the following equation:

$$F_0 \leq S_0 e^{(r+u)T}$$

This relationship differs from other types of futures contracts because, for a consumption asset, the firms holding these assets will generally use them as part of their operations. As a result, arbitrage opportunities are not necessarily seized, which is moreover contrary to one of the four assumptions stated. This is why we have an inequality rather than an equality, unlike other models.

Holding the asset allows, for example, the continuity of production, hence the idea of a yield associated with it. The measure of this, referred to in English as the "convenience yield," allows the two sides of the inequation to be balanced to reach an equality. It is, in a sense, a reflection of market expectations regarding the future availability of the commodity. If this rate is known at contract initiation, then:

$$S_0 e^{(r+u)T} = F_0 e^{uT} \Leftrightarrow F_0 = S_0 e^{(r+u-y)T} \quad (1)$$

If we denote by c the cost of carry, with $c = r + u$, we finally have:

$$F_0 = S_0 e^{(c-y)T} \quad (2)$$

Equation (2) will be the first building block for our modeling, allowing us to isolate and calculate the convenience yield over our data sample.

Measuring Geopolitical Risk, D. Caldara & M. Iacoviello.

Incorporating a variable capturing global geopolitical dynamics is intuitively relevant. This approach, however, requires a rigorous quantification of this risk — a need addressed precisely by the "Geopolitical Risk" (GPR) index developed by Caldara and Iacoviello (2022).

The introduction of this measure follows from the absence of an indicator that is consistent over time and able to incorporate the geopolitical component. Geopolitical risk is defined by the two authors as "the threat, realization, and escalation of adverse events associated with wars, terrorism, and any tensions among states and political actors that affect the peaceful course of international relations." Subsequently, attention is drawn to the inclusion of terrorism in this definition since, as noted, terrorist attacks have sometimes generated political tensions and appear to be a non-negligible variable that has intensified in recent times. This definition leads to the creation of two different indices: the geopolitical risk index and the geopolitical threat index. The GPR index captures both threats and realizations of adverse events, which is why we choose it for our analysis.

The conclusions drawn by the authors are unsurprising: higher-than-normal values of the GPR index are associated with a higher probability of economic disaster and lower expected GDP growth. A study conducted between 1985 and 2019 showed that a shock to the index induces a persistent decline in investment, the labor market, and stock prices.

This index nevertheless has a few limitations worth knowing before going further. By construction, it is subject to a definition of geopolitical risk that differs from the authors' own definition, since the index is based on newspapers that use these terms somewhat differently. Indeed, since it reflects the count of articles referring to geopolitical events and associated threats, the definition of each term follows journalistic practice rather than an exact definition.

A second significant limitation of this index is that it is built around ten English-language newspapers only. However, a comparison of the index between non-U.S. and U.S. newspapers indicates a correlation of 0.88, suggesting that geopolitical events are generally covered in a similar manner and that this limitation has little impact.

It should be kept in mind that this index reflects an estimate of the level of geopolitical risk (or threat) obtained through the use of a constructed "dictionary" used for word counting.

The addition of the VIX index is justified by the argument that the GPR index captures events that likely do not have an economic origin but could nevertheless increase during periods of high financial volatility and political uncertainty — an argument directly put forward by the authors. Thus, combining these two measures allows us to take into account more relevant information for our modeling.

The Role of Storage in Commodity Markets: Indirect Inference Based on Grains Data, C. Gouel & N. Legrand.

With a view to determining the storage cost, we will rely on the work of C. Gouel and N. Legrand (2022). The major contribution of their article concerns the structural estimation of storage costs.

The authors decompose the storage cost into two components: a physical unit cost and a rate of inventory deterioration. Their empirical study yields a deterioration rate of zero. This means that storage costs are entirely determined by the unit cost. This result is consistent with those of Cafiero et al. (2011).

Nevertheless, using the Monte Carlo method, the authors show that the model is unable to distinguish between unit costs and losses. These unit costs are estimated at 3.7% of the steady-state price using a numerical approximation method known as the "four-node" method. This price corresponds to the long-run equilibrium path of the market in the case where supply perfectly equals demand.

Adding the opportunity cost related to the real interest rate and the implicit cost, we have, in the steady-state case, the following equality:

$$k + 1 - \delta(1 - \delta)e^{g_p} = 0$$

Where:

- k: the physical cost

- δ : the discount factor
- $(1-\delta)$: the deterioration rate

e^{g_p} : the price trend factor

For simplification purposes, we will assume that the steady-state price used for our modeling is the average price of our database. Given that our data sample largely covers crisis periods, this assumption allows us to reflect a steady-state price relative to this particular period.

As this variable is fixed over time, whatever its value, its impact will remain minimal. It is more a matter of studying how we will incorporate it into our study.

Empirical Study

Model Definition.

Let us recall the equation defining the valuation of futures contracts whose underlying is a consumption asset:

$$F_0 = S_0 e^{(c-y)T} \quad (2)$$

Where F_0 denotes the futures price, S_0 the spot price at the initial date, c the cost of carry, y the convenience yield, and T the contract's maturity.

Unlike the standard modeling in which the storage cost is expressed as a proportion of the underlying's value, we assumed a fixed storage cost. This cost, denoted U , is therefore not directly usable as a rate in a continuous-time expression.

In the absence of arbitrage opportunities, two self-financed and risk-free strategies offering identical payoffs at maturity T must, by definition, have the same value at the initial date.

Based on this and on the assumptions of the stated model, we can establish the following strategy:

- At $t = 0$ (initial instant), the investor buys a quantity of the underlying (in this case, crude oil) at price S_0 , and takes a short position in a futures contract on this underlying with a strike price of F_0 .

- At maturity, the seller of the contract will have paid the storage costs, which we have treated as a fixed cost. However, they will have benefited from the convenience yield, which they will deduct from the contract, since the long position could not benefit from it.

By applying the principle of no-arbitrage and by compounding at the risk-free rate, we can establish the following relationship:

$$F_0 = (S_0 + U)e^{(r-y)T} \quad (3)$$

This expression reflects the formation of the futures price when the storage cost is treated as an absolute fixed cost rather than as a proportional rate, and allows the convenience yield to be isolated from observable quantities.

We obtain a hybrid formula between the two propositions (continuous time and discrete time) of J. Hull. This formula will allow us to incorporate our storage costs as defined while applying a convenience yield proportional to the price. Isolating this yield, we now have:

$$\begin{aligned} F_0/(S_0 + U) &= e^{(r-y)T} \Leftrightarrow \ln(F_0/(S_0 + U)) = (r - y)T \Leftrightarrow r - y \\ &= \frac{1}{T} \ln(F_0/(S_0 + U)) \Leftrightarrow y = r - \frac{1}{T} \ln(F_0/(S_0 + U)) \end{aligned} \quad (4)$$

Subject to the condition that:

- $S_0 + U$ is strictly nonzero
- $S_0 + U$ and S_0 have the same sign
- T is strictly nonzero

As mentioned in our introduction, we must now model this variable as a function of the GPR index and the VIX over time. This gives us the following equation:

$$y_t = \beta_0 + \beta_1 GPR_t + \beta_2 VIX_t + \varepsilon_t \quad (5)$$

Data Collection.

The data used cover precisely the period from 04/07/2014 to 04/07/2024, at daily frequency, for a total of 3,653 observations. The choice of this time interval is justified on several grounds. First, capturing ten years of daily data provides us with a sample large enough for our modeling. Furthermore, this time span reflects both periods of calm in financial markets as well as periods of crisis and major shocks. We can notably cite the collapse of oil prices at the start of the period following the

emergence of shale oil in the United States; of course, the Covid-19 crisis, whose impact was certainly not limited to oil but affected the global economy as a whole; and more recently, the war in Ukraine and, more broadly, rising geopolitical tensions around the world.

This choice of time window therefore allows us to reflect oil markets under different, more or less volatile, states.

Moreover, we have here a sufficiently recent data sample to reflect the markets more concretely than if we had used older data.

The spot and futures prices of WTI come from the U.S. Energy Information Administration (EIA). The Geopolitical Risk (GPR) index is available on M. Iacoviello's personal website. The VIX index is extracted from the FRED database of the Federal Reserve Bank of St. Louis, as are the risk-free rates derived from one-month U.S. Treasury bills.

As for the storage cost of the underlying (WTI), we assume the steady-state price to be the average price of our sample, namely \$63.27/bbl. Then, based on the work of C. Gouel and N. Legrand, we set the storage cost at 3.7% of this price. Therefore: $U = \$2.341/\text{bbl}$.

Stationarity of the Data.

The stationarity of the series is a fundamental prerequisite prior to our econometric modeling. To verify this condition, we use the Augmented Dickey-Fuller (ADF) test, whose null hypothesis posits the presence of a unit root implying non-stationarity. The results, reported in Table 1, lead to the rejection of this null hypothesis (H_0) for all three series at the 5% significance level: the test statistics are indeed well below the critical value of -2.86, allowing us to conclude that each of our variables is stationary. To be even more thorough, since the critical value at the 1% threshold is -3.43, all of our series are also stationary at that level.

Table 1. Results of the Augmented Dickey-Fuller Test.

Series	VIX Index	GPR Index	Convenience Yield
Test statistic	-6.477	-30.841	-17.033
Critical value (5% level)	-2.86	-2.86	-2.86
Stationarity	Yes	Yes	Yes

Furthermore, examining the correlation between our two exogenous variables, we obtain: $\rho_{\text{VIX,GPR}} = 1.687\%$. This value is low but nonzero. It remains too weak,

however, to indicate a real causal link between the two exogenous variables of our study.

Final Results.

The regression of our model yields an R^2 of about 2.53%, meaning that the model explains approximately 2.5% of the variability of the convenience yield. As for the adjusted R^2 , denoted $\overline{R}^2$, we have:

$$\overline{R}^2 = 1 - (1 - R^2)(n - 1)(n - k - 1)$$

In our case, we therefore have:

$$\begin{aligned}\overline{R}^2 &= 1 - (1 - R^2)(n - 1)/(n - k - 1) = 1 - (1 - 2.5259\%)(3654 \\ &\quad - 1)/(3654 - 2 - 1) = 1 - (1 - 2.5259\%) * 1.0005748 \\ &\approx 2.4725\%\end{aligned}$$

We therefore have an adjusted R^2 of around 2.47%, a difference of 0.06%. The adjusted R^2 corrects the bias of the standard R^2 by introducing a penalty for each degree of freedom lost. This makes it possible to assess whether adding a variable improves the model more than chance alone would allow. This small gap between the two is unsurprising given the number of observations that make up our sample.

This relatively low value explains why the standard error of the residuals is relatively high. This measure represents the dispersion of data points around the regression line.

$$SE = \sqrt{SSR/(n - k - 1)} = \sqrt{130.38/(3654 - 3)} \approx 18.897\%$$

Table 2. Regression Statistics.

Statistic	Value
Multiple correlation coefficient	15.893%
R^2	2.5259%
Adjusted R^2	2.4725%
Standard error of residuals	18.897%
Observations	3654

The Fisher statistic, which assesses the overall relevance of the model, is based on the following formula:

$$F = (R^2/k)/[(1 - R^2)/(n - k - 1)] \\ = (2.5259\%/2)/[(1 - 2.5259\%)/(3654 - 2 - 1)] \approx 47.36$$

The interpretation relies on the following hypothesis test:

- Null hypothesis (H0): the model has no explanatory power; the endogenous variable is not influenced by any of the independent variables.
- Alternative hypothesis (H1): at least one of the coefficients β_i ($i = 1,2$) is significantly different from zero, meaning the model has overall usefulness.

The decision rule requires comparing the empirical statistic to the critical F value from the table (based on the error tolerance threshold and the degrees of freedom). In our case:

- At the 5% level: $F_{\text{critical}} \approx 2.995$.
- At the 1% level: $F_{\text{critical}} \approx 4.605$.

Thus, regardless of the tolerance threshold considered, our model is significant, since our empirical statistic exceeds the critical value in both cases.

The above formula simply illustrates that the higher R^2 is, the closer $1 - R^2$ tends to zero, and therefore the higher the Fisher statistic becomes.

We are now assured that our model is statistically significant despite its low explanatory power.

"When a result is described as 'significant,' one should not conclude that the effect it describes is powerful. It simply means that it is unlikely that the observed result is the product of chance" (D. Kahneman, O. Sibony & C.-R. Sunstein, Noise: A Flaw in Human Judgment).

Analysis of the Standard Error of Residuals

The standard error of the residuals measures the average dispersion of the observed values of the convenience yield around the values predicted by the model. Concretely, this means that our model "errs" on average by about 19% in its prediction. The combination of a low R^2 and a high residual error is consistent and reveals that the VIX and the GPR together explain only about 2.5% of the endogenous variable, with the rest being noise captured by the residuals. The standard error is therefore mainly driven by factors not captured by our variables.

The ratio $\text{residuals}/\text{total} = 130.38 / 133.759 \approx 97.5\%$ confirms that the vast majority of the variability of the convenience yield escapes the model. To make sure we are

not facing a spurious correlation, we must verify the stationarity of the residuals. For this, we again perform an ADF test, with:

- Null hypothesis: the residuals have a unit root (non-stationary)
- Alternative hypothesis: the residuals are stationary.

The results (see Table 3) clearly show that we have stationarity at the 5% critical threshold, and even at the 1% level, since the critical value in that case is -3.34. In both cases we fall below the critical value.

Table 3. ADF Test of the Residuals.

	Value
Test statistic	-17.78
Critical value (5%)	-2.86
Stationarity	Yes

ANOVA Analysis

The total variance of the convenience yield is 133.759. The VIX and the GPR explain only 3.3786 of it, i.e., about 2.53%. The remaining 130.38 (97.47%) is attributable to factors outside the model. This is the direct translation of the 2.526% R^2 of our regression.

On average, the mean variance explained by each exogenous variable is: $3.3786/2 = 1.689$.

The P-value shown as zero means it is below the numerical precision threshold of the software. This observation means that the probability of observing the value of the Fisher statistic under the null hypothesis of non-significance of the model is practically nil.

Table 4. ANOVA.

	Degrees of freedom	Sum of squares	Mean square	F-Value	P-Value
Regression	2	3.3786	1.689	47.305	0
Residuals	3651	130.38	0.0357	—	—
Total	3653	133.759	—	—	—

The model's constant represents the structural level of the convenience yield when both risk indices are zero. It stands at 0.4932, with a 95% confidence interval

between [0.4729; 0.513]. This interval is relatively narrow, which reflects a certain precision, also visible in the low standard error of about 1%. Statistically, with a T-stat of 47.783 and a P-value of zero, the constant is highly significant.

The coefficient associated with the VIX index stands at 0.2555%, with a 95% confidence interval between [0.171%; 0.34%], thus excluding the value zero. The gap between these two bounds remains contained, which is consistent with a fairly low standard error. Moreover, this coefficient is statistically significant, with a T-stat of 5.92 (critical threshold at 1.96) and a P-value close to zero.

Finally, the coefficient associated with the GPR index is relatively small and negative, with a 95% confidence interval of [-0.000578; -0.000347], also excluding the value zero. Despite this, it is a statistically explanatory coefficient, with a T-stat well above 1.96 in absolute value and a nearly zero P-value.

Overall, the results in Table 5 confirm that all estimated coefficients are statistically significant, regardless of their respective magnitude. Furthermore, the 95% confidence intervals for each coefficient systematically exclude the value zero, confirming their significance.

Table 5. Regression Coefficient Results.

	Coefficients	Standard Error	T-Stat	P-Value	Lower 95%	Upper 95%
Constant	0.4932	0.0103	47.783	0	0.4729	0.513
VIX Index	0.2555%	0.0431958%	5.91566	3.60961×10^{-9}	0.00171	0.0034
GPR Index	-0.0462%	5.91323×10^{-9}	-7.8198	6.82×10^{-15}	-0.000578	-0.000347

We therefore have a solid foundation in terms of statistical significance, despite a weak explanatory power, on which we can now take a critical look at the relationships identified and the limitations of our study.

Discussion of Results

Interpretation of the Final Model.

The results of our study lead us to the following model:

$$y_t = 0.4932 - 0.0462\% * GPR_t + 0.2555\% * VIX_t \quad (5)$$

The constant represents the structural equilibrium level of the convenience yield when both risk indices are simultaneously zero. It stands at 0.4932. This level reflects the existence of a structural premium, independent of any cyclical conditions. Although it is highly significant from a statistical standpoint, its literal interpretation has no observable empirical reality, since the indices are never simultaneously zero. In particular, the VIX index, which measures the implied volatility of S&P 100 options, has historically never reached zero; its floor seems instead to sit around 10 to 12 points. Likewise, the GPR index, which reflects media coverage of global geopolitical tensions, has only an infinitesimal probability of being zero.

Thus, the constant is more akin to an adjustment parameter than an interpretable economic quantity.

The negative coefficient associated with the GPR index constitutes a counter-intuitive result of the model. It reveals that geopolitical risk does not act as a catalyst for the convenience yield. Concretely, since it is an indicator of geopolitical sentiment, the natural intuition would be that any rise in geopolitical risk, and therefore in the index, should raise the value of physically holding the asset and thus increase the convenience yield. Several hypotheses emerge to account for this inconsistency.

It is possible that, in periods of confirmed geopolitical tension, economic agents redirect their portfolios toward more "classic" safe-haven assets such as gold or sovereign bonds. This diverts demand away from the physical asset in question, reducing the utility of holding it immediately. Geopolitical risk may therefore act as a signal for reallocation rather than as a signal for physical precaution.

Geopolitical risk can potentially induce expectations of a lasting economic slowdown. If operators anticipate this type of contraction, future demand for the physical asset will be revised downward. The utility of holding the asset, which depends on anticipated future demand, therefore decreases as a result. As discussed in the introduction, D. Caldara and M. Iacoviello showed that a rise in their index induces lower expected GDP growth. Moreover, a study conducted between 1985 and 2019 showed that a shock to the index induces a persistent decline in investment, the labor market, and stock prices.

Geopolitical tensions can also be accompanied by increased logistical frictions (insurance premiums, disruptions to transport routes, additional costs of securing

inventories, etc.). These additional costs could directly affect the benefit associated with holding the asset.

Certain geopolitical episodes can trigger public interventions in the form of releases from national strategic reserves. Such events can create a temporary decline in the premium offered for holding this underlying asset.

Despite all these hypotheses, it should be recalled that the impact of this exogenous variable remains minor relative to the variability of the endogenous variable, with the average explained variance being about 1.689% per variable.

The VIX index, which measures the implied volatility of S&P 100 options, reflects expectations of future financial market volatility rather than realized volatility. It is generally interpreted as a "fear gauge." It is associated with a positive coefficient, which seems intuitive. A rise in the VIX signals intensifying uncertainty, prompting economic agents to secure their supplies, notably through physical inventories, which translates into a higher holding premium.

This nevertheless assumes that the impact of the VIX is linear. However, it is possible that this effect is not linear: beyond a certain threshold, the transmission mechanisms may become saturated, or conversely, may intensify.

Its dynamics differ from those of the GPR index, which is an aggregated indicator of geopolitical sentiment.

Decomposition of Explained Variability.

In order to more deeply identify the dynamics driving the convenience yield, it is worth analyzing the decomposition of the variability explained by our model. To do so, we will use the partial R^2 method as well as the semi-partial R^2 method.

The first approach is the simplest; it consists of measuring the marginal contribution of each variable, i.e., the additional variance explained by each exogenous variable.

$$R_{partial,VIX}^2 = \frac{t_{VIX}^2}{t_{VIX}^2 + df(residual)} = 5.92^2 / (5.92^2 + 3651) \approx 0.949\%$$

$$R_{partial,GPR}^2 = \frac{t_{GPR}^2}{(t_{GPR}^2 + df(residual))} = (-7.82)^2 / ((-7.82)^2 + 3651) \approx 1.652\%$$

The second method measures the absolute contribution of each variable to the total R^2 .

$$\begin{aligned} R_{semipartial,VIX}^2 &= (t_{VIX}/\sqrt{n-k-1})^2 * (1 - R^2) \\ &= 5.92^2/3651 * (1 - 0.025259) \approx 0.935\% \end{aligned}$$

$$\begin{aligned} R_{semipartial,GPR}^2 &= (t_{GPR}/\sqrt{n-k-1})^2 * (1 - R^2) \\ &= (-7.82)^2/3651 * (1 - 0.02529) \approx 1.633\% \end{aligned}$$

Whichever method is used, the hierarchy is the same: the GPR index explains more than the VIX index. This is also what the higher T-stat of the first index compared to the second suggested. The GPR index explains about 1.74 times more than the VIX index. This also means that between 60% and 70% of our model's predictive power rests exclusively on the GPR index, which therefore largely dominates the economic and financial volatility index.

Interpretation of the Dynamics.

From this, we have been able to establish a hierarchy in the contribution of the exogenous variables, as well as some possible explanations for understanding their impact. What we can take away here is that the benefit granted to the physical holding of oil (WTI) relative to a futures contract depends more on geopolitical uncertainty than on the index measuring economic uncertainty. While this seems fairly intuitive, the impact of these variables is not.

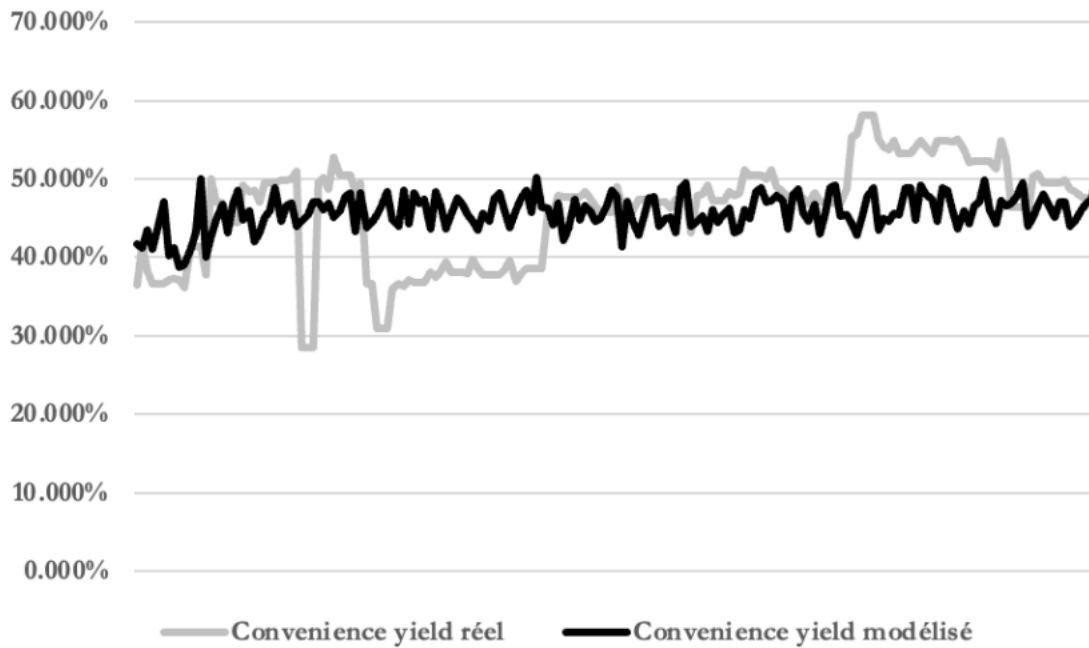
The influence of the VIX is entirely consistent, since we concluded that economic instability positively impacts our dependent variable. As for the geopolitical instability index, we observed that it takes precedence over the other explanatory element and that it is negatively correlated with it. Since it is never zero, this means it exerts a permanent structural downward pressure.

Fundamentally, this means that the market distinguishes between the two types of uncertainty: financial/economic uncertainty, captured by the VIX, increases the value of holding the asset; whereas geopolitical uncertainty, captured by the GPR, decreases the value of holding it.

If one can assume that the price of the underlying may be strongly affected by these dynamics, it appears that the convenience yield itself is much less affected by these factors.

On the other hand, with an R^2 of around 2.5%, overall risk — whether financial, economic, or geopolitical — only marginally explains the variance of the convenience yield, whose dynamics seem more complex to explain with only our variables.

Figure 1. Representation of the modeled and actual convenience yield between 10/07/2023 and 04/07/2024



It is therefore possible that the dynamics affecting our endogenous variable relate more to factors linked to storage capacity, to the futures market itself, and to physical-market participants (refiners, oil companies, traders, etc.).

Indeed, since the convenience yield reflects, in a sense, expectations regarding the availability of the underlying, it seems likely that it depends mainly on the factors linked to that.

Limitations of the Model.

It is important to approach our model with a critical eye by examining its possible limitations.

First, we assumed that $S_0 + U$ and F_0 had the same sign. What is this in practice? This assumption seems acceptable since, in our data sample, this situation turns out

not to hold on only one occasion. We can therefore consider that it has no real impact on the results of our study.

Next, the futures valuation model implies several assumptions. First, the absence of transaction costs, which is not true in practice. The same goes for the possibility of borrowing and lending at the same risk-free rate. Finally, the tax rate on profits is not actually fixed regardless of the amount.

A strong assumption we made concerns the value of storage costs. Although we relied on the work of C. Gouel and N. Legrand, assuming these costs to be fixed is a central and restrictive assumption. Looking more closely, it appears that these costs could be broken down into two distinct parts: a fixed part that applies regardless of the contract price, and a variable part that does indeed depend on the price of the underlying.

Treating this variable as a fixed cost may therefore affect our model, since it reduces its significance by not being a time-dependent variable.

Moreover, the work we relied on suggests using a fixed percentage of the steady-state price. However, this poses an identification problem, since the steady-state price is not directly observable. We assumed that this price was the average of our sample.

Furthermore, the time period chosen for our data sample, chosen a priori to reflect both times of crisis and calmer periods, may also carry a bias related to its specificity. In addition, the very measure of the convenience yield, which results from a calculation rather than a direct data point, may be subject to data imperfections, for example maturity mismatches or certain variables that do not share the same calendar. Moreover, since the data is daily, it is likely to contain a great deal of noise.

We might have to limit our conclusions to dynamics that only apply in times of uncertainty and crisis, which is broadly what our data sample reflects.

Figure 1 allows us to visually appreciate, over the last six months of our sample, the changes between the modeled premium and the actual premium. It appears that the representation of the modeled convenience yield is much more stable, and less subject to large variations, than the actual variable. Calculating the standard deviation of each, we find that:

- The standard deviation of the actual variable is 19.14%
- The standard deviation of the modeled variable is 3.04%

This aligns with our conclusion that we only explain a small part of the variability of the actual variable.

The use of the ordinary least squares method naturally assumes a linear relationship between our two exogenous variables and the endogenous variable. However, the oil holding premium may exhibit asymmetric and nonlinear behavior.

Some Avenues for Reflection.

With a view to deepening our understanding of the dynamics explaining the premium offered for holding oil, we can mention several possible avenues for reflection that could increase the explained variability (R^2).

First, the study of the "roll yield," namely the gain or loss realized when "rolling" a futures position approaching maturity into a further-dated one, could be of importance.

Moreover, examining the effects on the dynamics of certain sectors related to our endogenous variable could prove relevant. Indeed, in their work, M. Iacoviello and D. Caldara raise the idea that a firm will gradually reduce its investments over the two quarters following a geopolitical shock. Thus, studying the impact of the GPR index over time could be significant.

The works of Kaldor (1939), Working (1949), and Brennan (1958) raise the idea that agents are willing to pay a premium to physically hold an asset today rather than in the future, and that this premium intensifies as overall uncertainty rises. Their conclusions explain that this is a persistent mechanism regardless of the situation.

They therefore consider the convenience yield to be a function of uncertainty, unlike Keynes (1930) and Hicks (1939), for whom this variable is a consequence of the market's hedging structure rather than a holding premium in the strict sense.

Buyers of futures contracts are largely agents with a hedging need. However, few market participants need to hedge crude oil itself, but rather fuels, which may divert them from this type of contract toward contracts more specific to their needs.

Conclusion.

The objective of this work is to identify the dynamics that affect the premium offered for holding oil, not to build a predictive model. Furthermore, it appears that the explanatory power of uncertainty remains marginal in identifying these dynamics,

since we are unable to explain more than 2.5% of the variability of this premium. This result may seem surprising, but one must distinguish the price of the underlying itself from the benefit associated with directly holding it.

In addition, we have attempted to identify the possible limitations of our study and our model that could precisely undermine this identification objective, while proposing avenues for research that could further enrich this work.

We are nonetheless certain that our model is statistically significant, which means that, despite the weakness of our model in terms of identifying the dynamics, this is by no means due to chance.

Furthermore, we succeeded in highlighting the fact that the geopolitical risk index is the dominant factor in explaining the endogenous variable relative to the VIX index. Although this conclusion seems intuitive, it has thus been statistically verified.

The works of Kaldor (1939), Working (1949), and Brennan (1958) suggest that the convenience yield could reflect physical scarcity. Thus, studying inventory levels over time proves to be a key point that escapes our modeling. For example, if a war triggers panic in the markets, but strategic reserves and global inventories are high, the holding premium has no reason to rise immediately.

Finally, we were able to identify the direction in which the various uncertainty dynamics studied can affect the convenience yield — namely, negatively for geopolitics and positively for the economy.

References / Bibliography

- Caldara, D., & Iacoviello, M. (2022). Measuring Geopolitical Risk. *American Economic Review*, 112(4), 1195-1225.
- Gouel, C., & Legrand, N. (2022). The Role of Storage in Commodity Markets: Indirect Inferences Based on Grains Data. CEPII Working Paper, No 2022-04.
- Hull, J.C. (2021). *Options, Futures, and Other Derivatives* (11th ed.). Pearson.
- Froot, K. A., Scharfstein, D. S., & Stein, J. C. (1993). Risk Management: Coordinating Corporate Investment and Financing Policies. *The Journal of Finance*, 48(4), 1629-1658.

- Brennan, M. J. (1958). The Supply of Storage. The American Economic Review, 48(1), 50-72.
- Hicks, J. R. (1939). Value and Capital: An Inquiry into Some Fundamental Principles of Economic Theory. Clarendon Press.
- Kaldor, N. (1939). Speculation and Economic Stability. The Review of Economic Studies, 7(1), 1-27.
- Keynes, J.M. (1930). A Treatise on Money (Vol. 2: The Applied Theory of Money).
- Working, H. (1949). The Theory of Price of Storage. The American Economic Review, 39(6), 1254-1262.